Sets

Set - Definition

A set is an unordered collection of different elements. A set can be written explicitly by listing its elements using set bracket. If the order of the elements is changed or any element of a set is repeated, it does not make any changes in the set.

Some Example of Sets

- A set of all positive integers
- A set of all the planets in the solar system
- A set of all the states in India
- A set of all the lowercase letters of the alphabet

Representation of a Set

Sets can be represented in two ways −

- Roster or Tabular Form
- Set Builder Notation

Roster or Tabular Form

The set is represented by listing all the elements comprising it. The elements are enclosed within braces and separated by commas.

Example 1 − Set of vowels in English alphabet, $A=\{a,e,i,o,u\}$A={a,e,i,o,u}
Example 2 − Set of odd numbers less than 10, $B=\{1,3,5,7,9\}$B={1,3,5,7,9}

Set Builder Notation

The set is defined by specifying a property that elements of the set have in common. The set is described as $A=\{x:p(x)\}$A={x:p(x)}
Example 1 − The set $\{a,e,i,o,u\}$\{a,e,i,o,u\} is written as −
$A=\{x:x \text{ is a vowel in English alphabet}\}$A={x:x is a vowel in English alphabet}
Example 2 − The set $\{1,3,5,7,9\}$\{1,3,5,7,9\} is written as −
$B=\{x:1\leq x<10 \text{ and } (x\%2)\neq 0\}$B={x:1≤x<10 and (x%2)≠0}
If an element x is a member of any set S, it is denoted by $x\in S$x∈S and if an element y is not a member of set S, it is denoted by $y\notin S$y∉S.
Example − If $S=\{1,1.2,1.7,2\},1\in S$S={1,1.2,1.7,2},1∈S but $1.5\notin S$1.5∉S

Some Important Sets

N − the set of all natural numbers = $\{1,2,3,4,.....\}$\{1,2,3,4,.....\}
Z − the set of all integers = $\{.....,-3,-2,-1,0,1,2,3,.....\}$\{.....,−3,−2,−1,0,1,2,3,.....\}

Z^{+} − the set of all positive integers

Q − the set of all rational numbers

R − the set of all real numbers

W − the set of all whole numbers

Cardinality of a Set

Cardinality of a set S, denoted by $|S|$|S|, is the number of elements of the set. The number is also referred as the cardinal number. If a set has an infinite number of elements, its cardinality is ∞∞.

Example − $|\{1,4,3,5\}|=4,|\{1,2,3,4,5,\ldots\}|=\infty$|{1,4,3,5}|=4,|{1,2,3,4,5,…}|=∞

If there are two sets X and Y,

- $|X|=|Y|$|X|=|Y| denotes two sets X and Y having same cardinality. It occurs when the number of elements in X is exactly equal to the number of elements in Y. In this case, there exists a bijective function 'f' from X to Y.
- $|X|\leq|Y|$|X|≤|Y| denotes that set X's cardinality is less than or equal to set Y's cardinality. It occurs when number of elements in X is less than or equal to that of Y. Here, there exists an injective function 'f' from X to Y.
- $|X|<|Y|$|X|<|Y| denotes that set X's cardinality is less than set Y's cardinality. It occurs when number of elements in X is less than that of Y. Here, the function 'f' from X to Y is injective function but not bijective.
- If $|X|\leq|Y|$If |X|≤|Y| and $|X|\geq|Y|$|X|≥|Y| then $|X|=|Y|$|X|=|Y|. The sets X and Y are commonly referred as equivalent sets.

Types of Sets

Sets can be classified into many types. Some of which are finite, infinite, subset, universal, proper, singleton set, etc.

Finite Set

A set which contains a definite number of elements is called a finite set.

Example − $S=\{x|x\in N$S={x|x∈N and $70>x>50\}$70>x>50}

Infinite Set

A set which contains infinite number of elements is called an infinite set.

Example − $S=\{x|x\in N$S={x|x∈N and $x>10\}$x>10}

Subset

A set X is a subset of set Y (Written as $X\subseteq Y$X⊆Y) if every element of X is an element of set Y.

Example 1 − Let, $X=\{1,2,3,4,5,6\}$ and $Y=\{1,2\}$. Here set Y is a subset of set X as all the elements of set Y is in set X. Hence, we can write $Y \subseteq X$.

Example 2 − Let, $X=\{1,2,3\}$ and $Y=\{1,2,3\}$. Here set Y is a subset (Not a proper subset) of set X as all the elements of set Y is in set X. Hence, we can write $Y \subseteq X$.

Proper Subset

The term "proper subset" can be defined as "subset of but not equal to". A Set X is a proper subset of set Y (Written as $X \subset Y$) if every element of X is an element of set Y and $|X|<|Y|$.

Example − Let, $X=\{1,2,3,4,5,6\}$ and $Y=\{1,2\}$. Here set $Y \subset X$ since all elements in Y are contained in X too and X has at least one element is more than set Y.

Universal Set

It is a collection of all elements in a particular context or application. All the sets in that context or application are essentially subsets of this universal set. Universal sets are represented as U.

Example − We may define U as the set of all animals on earth. In this case, set of all mammals is a subset of U, set of all fishes is a subset of U, set of all insects is a subset of U, and so on.

Empty Set or Null Set

An empty set contains no elements. It is denoted by $\emptyset$. As the number of elements in an empty set is finite, empty set is a finite set. The cardinality of empty set or null set is zero.

Example − $S=\{x|x \in N$ and $7<x<8\}=\emptyset$

Singleton Set or Unit Set

Singleton set or unit set contains only one element. A singleton set is denoted by $\{s\}$.

Example − $S=\{x|x \in N, 7<x<9\} = \{8\}$

Equal Set

If two sets contain the same elements they are said to be equal.

Example − If $A=\{1,2,6\}$ and $B=\{6,1,2\}$, they are equal as every element of set A is an element of set B and every element of set B is an element of set A.

Equivalent Set

If the cardinalities of two sets are same, they are called equivalent sets.

Example − If $A=\{1,2,6\}$ and $B=\{16,17,22\}$, they are equivalent as cardinality of A is equal to the cardinality of B. i.e. $|A|=|B|=3$

Overlapping Set

Two sets that have at least one common element are called overlapping sets.

In case of overlapping sets −

- $n(A\cup B)=n(A)+n(B)-n(A\cap B)$n(A∪B)=n(A)+n(B)−n(A∩B)
- $n(A\cup B)=n(A-B)+n(B-A)+n(A\cap B)$n(A∪B)=n(A−B)+n(B−A)+n(A∩B)
- $n(A)=n(A-B)+n(A\cap B)$n(A)=n(A−B)+n(A∩B)
- $n(B)=n(B-A)+n(A\cap B)$n(B)=n(B−A)+n(A∩B)

Example − Let, $A=\{1,2,6\}$A={1,2,6} and $B=\{6,12,42\}$B={6,12,42}. There is a common element '6', hence these sets are overlapping sets.

Disjoint Set

Two sets A and B are called disjoint sets if they do not have even one element in common. Therefore, disjoint sets have the following properties −

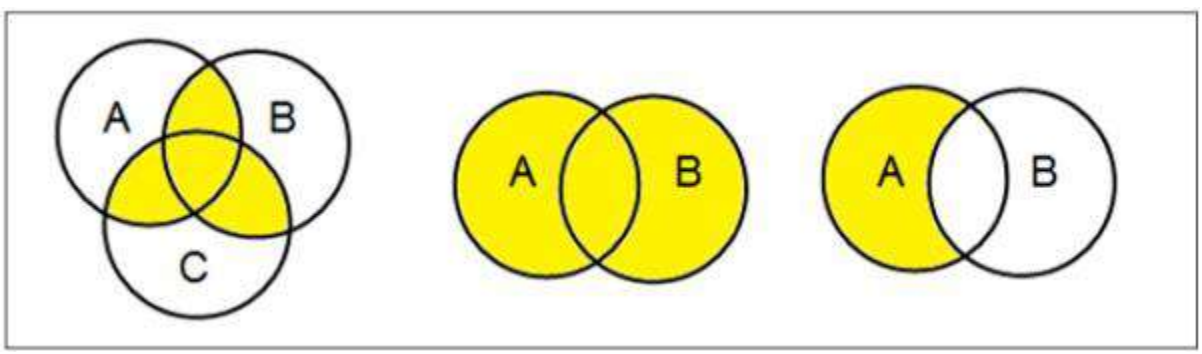
- $n(A\cap B)=\emptyset$n(A∩B)=∅
- $n(A\cup B)=n(A)+n(B)$n(A∪B)=n(A)+n(B)

Example − Let, $A=\{1,2,6\}$A={1,2,6} and $B=\{7,9,14\}$B={7,9,14}, there is not a single common element, hence these sets are overlapping sets.

Venn Diagrams

Venn diagram, invented in 1880 by John Venn, is a schematic diagram that shows all possible logical relations between different mathematical sets.

Examples



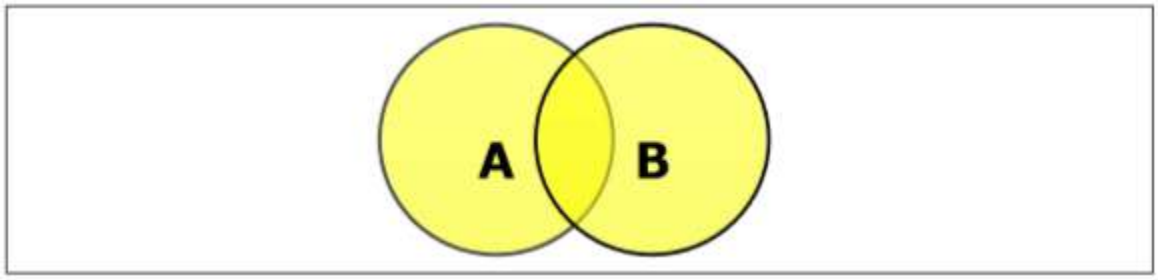
Set Operations

Set Operations include Set Union, Set Intersection, Set Difference, Complement of Set, and Cartesian Product.

Set Union

The union of sets A and B (denoted by $A\cup B$A∪B) is the set of elements which are in A, in B, or in both A and B. Hence, $A\cup B=\{x|x\in A \text{ OR } x\in B\}$A∪B={x|x∈A OR x∈B}.

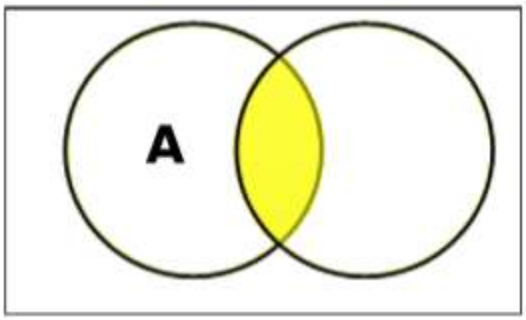
Example − If $A=\{10,11,12,13\}$ and $B=\{13,14,15\}$, then $A\cup B=\{10,11,12,13,14,15\}$. (The common element occurs only once)



Set Intersection

The intersection of sets A and B (denoted by $A\cap B$) is the set of elements which are in both A and B. Hence, $A\cap B=\{x|x\in A \text{ AND } x\in B\}$.

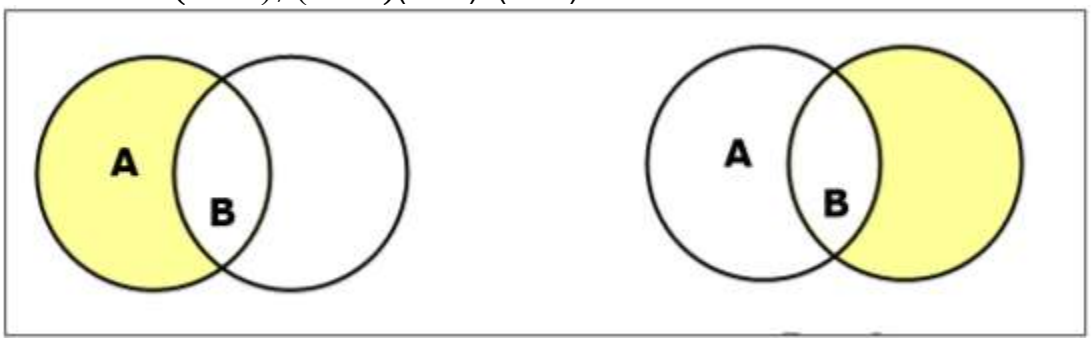
Example − If $A=\{11,12,13\}$ and $B=\{13,14,15\}$, then $A\cap B=\{13\}$.



Set Difference/ Relative Complement

The set difference of sets A and B (denoted by $A-B$) is the set of elements which are only in A but not in B. Hence, $A-B=\{x|x\in A \text{ AND } x\notin B\}$.

Example − If $A=\{10,11,12,13\}$ and $B=\{13,14,15\}$, then $(A-B)=\{10,11,12\}$ and $(B-A)=\{14,15\}$. Here, we can see $(A-B)\neq(B-A)$



Complement of a Set

The complement of a set A (denoted by A'A′) is the set of elements which are not in set A. Hence, $A'=\{x|x\notin A\}$A′={x|x∉A}.
More specifically, $A'=(U-A)$A′=(U−A) where UU is a universal set which contains all objects.

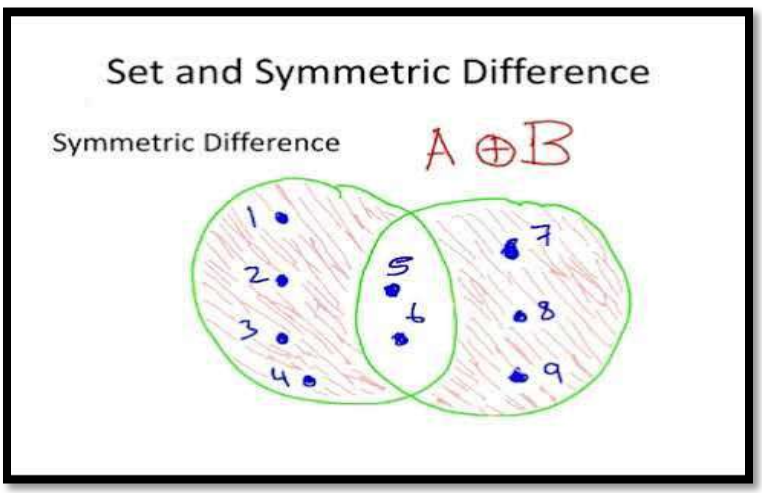
The symmetric difference: it is using Venn diagram of two subsets A and B is a sub set of U, denoted by A △ B and is defined by

A **△** B = (A – B) ∪ (B – A)

Let A and B are two sets. The symmetric difference of two sets A and B is the set (A – B) ∪ (B – A) and is denoted by A △ B.

Thus, A **△** B = (A – B) ∪ (B – A) = {x : x ∉ A ∩ B}

or, A **△** B = {x : [x ∈ A and x ∉ B] or [x ∈ B and x ∉ A]}



The shaded part of the given Venn diagram represents **A △ B**.

A △ B is the set of all those elements which belongs either to A or to B but not to both.

A △ B is also expressed by (A ∪ B) - (B ∩ A).

It follows that A △ ∅ = A for all subset A,

A △ A = ∅ for all subset A

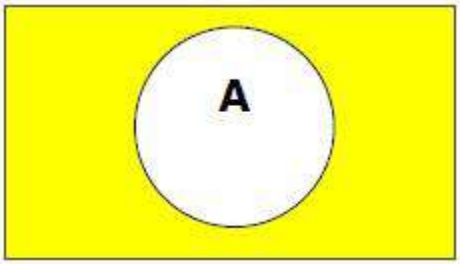
Properties of symmetric difference:

(i) A △ B = B △ A; [Commutative property]

(ii) A △ (B △ C) = (A △ B) △ C [Associative property]

Example −

If $A=\{x|x \text{ belongstosetofoddintegers}\}$A={x|x belongstosetofoddintegers} then $A'=\{y|y \text{ doesnotbelongtosetofoddintegers}\}$A′={y|y doesnotbelongtosetofoddintegers}



Cartesian Product / Cross Product

The Cartesian product of n number of sets $A_1,A_2,\ldots A_n$A1,A2,…An denoted as $A_1\times A_2\cdots\times A_n$A1×A2⋯×An can be defined as all possible ordered pairs $(x_1,x_2,\ldots x_n)$(x1,x2,…xn) where $x_1\in A_1,x_2\in A_2,\ldots x_n\in A_n$x1∈A1,x2∈A2,…xn∈An

Example − If we take two sets $A=\{a,b\}$A={a,b} and $B=\{1,2\}$B={1,2},

The Cartesian product of A and B is written as

− $A\times B=\{(a,1),(a,2),(b,1),(b,2)\}$A×B={(a,1),(a,2),(b,1),(b,2)}

The Cartesian product of B and A is written as

− $B\times A=\{(1,a),(1,b),(2,a),(2,b)\}$B×A={(1,a),(1,b),(2,a),(2,b)}

Power Set

Power set of a set S is the set of all subsets of S including the empty set. The cardinality of a power set of a set S of cardinality n is 2^n2n. Power set is denoted as $P(S)$P(S).

Example −

For a set $S=\{a,b,c,d\}$S={a,b,c,d} let us calculate the subsets −

- Subsets with 0 elements − $\{\emptyset\}${∅} (the empty set)
- Subsets with 1 element − $\{a\},\{b\},\{c\},\{d\}${a},{b},{c},{d}
- Subsets with 2 elements − $\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\}${a,b},{a,c},{a,d},{b,c},{b,d},{c,d}
- Subsets with 3 elements − $\{a,b,c\},\{a,b,d\},\{a,c,d\},\{b,c,d\}${a,b,c},{a,b,d},{a,c,d},{b,c,d}
- Subsets with 4 elements − $\{a,b,c,d\}${a,b,c,d}

Hence, $P(S)$=P(S)=

{{∅},{a},{b},{c},{d},{a,b},{a,c},{a,d},{b,c},{b,d},{c,d},{a,b,c},{a,b,d},{a,c,d},{b,c,d},{a,b,c,d}}{{∅},{a},{b},{c},{d},{a,b},{a,c},{a,d},{b,c},{b,d},{c,d},{a,b,c},{a,b,d},{a,c,d},{b,c,d},{a,b,c,d}}

$|P(S)|=2^4=16$|P(S)|=24=16

Note − The power set of an empty set is also an empty set.

$|P(\{\emptyset\})|=2^0=1$|P({∅})|=20=1

Algebra of Sets(,Set Identities)

Sets under the operations of union, intersection, and complement satisfy various laws (identities) which are listed in Table 1.

Table: Law of Algebra of Sets

Idempotent Laws	(a) $A \cup A = A$	(b) $A \cap A = A$
Associative Laws	(a) $(A \cup B) \cup C = A \cup (B \cup C)$	(b) $(A \cap B) \cap C = A \cap (B \cap C)$
Commutative Laws	(a) $A \cup B = B \cup A$	(b) $A \cap B = B \cap A$
Distributive Laws	(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	(b) $A \cap (B \cup C) =(A \cap B) \cup (A \cap C)$
De Morgan's Laws	(a) $(A \cup B)^c=A^c\cap B^c$	(b) $(A \cap B)^c=A^c\cup B^c$
Identity Laws	(a) $A \cup \emptyset = A$ (b) $A \cup U = U$	(c) $A \cap U =A$ (d) $A \cap \emptyset = \emptyset$
Complement Laws	(a) $A \cup A^c= U$ (b) $A \cap A^c= \emptyset$	(c) $U^c= \emptyset$ (d) $\emptyset^c = U$
Involution Law	(a) $(A^c)^c = A$	

Table 1 shows the law of algebra of sets.

Example 1: Prove Idempotent Laws:

1. (a) $A \cup A = A$

Solution:

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Since, B ⊂ A ∪ B, therefore A ⊂ A ∪ A
Let   x ∈ A ∪ A ⇒ x ∈ A  or   x ∈ A ⇒  x ∈ A
∴ A ∪ A ⊂ A
As  A ∪ A ⊂ A and  A ⊂ A ∪ A ⇒ A =A ∪ A. Hence Proved.
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Inclusion-Exclusion Principle

In combinatorics, a branch of mathematics, the **inclusion–exclusion principle** is a counting technique which generalizes the familiar method of obtaining the number of elements in the union of two finite sets; symbolically expressed as

Let A, B be any two finite sets. Then n (A ∪ B) = n (A) + n (B) - n (A ∩ B)

Here "include" n (A) and n (B) and we "exclude" n (A ∩ B)

Example 1:

Suppose A, B, C are finite sets. Then A ∪ B ∪ C is finite and n (A ∪ B ∪ C) = n(A) + n(B) + n(C) - n(A ∩ B) - n(A ∩ C) - n(B ∩ C) + n(A ∩ B ∩ C)

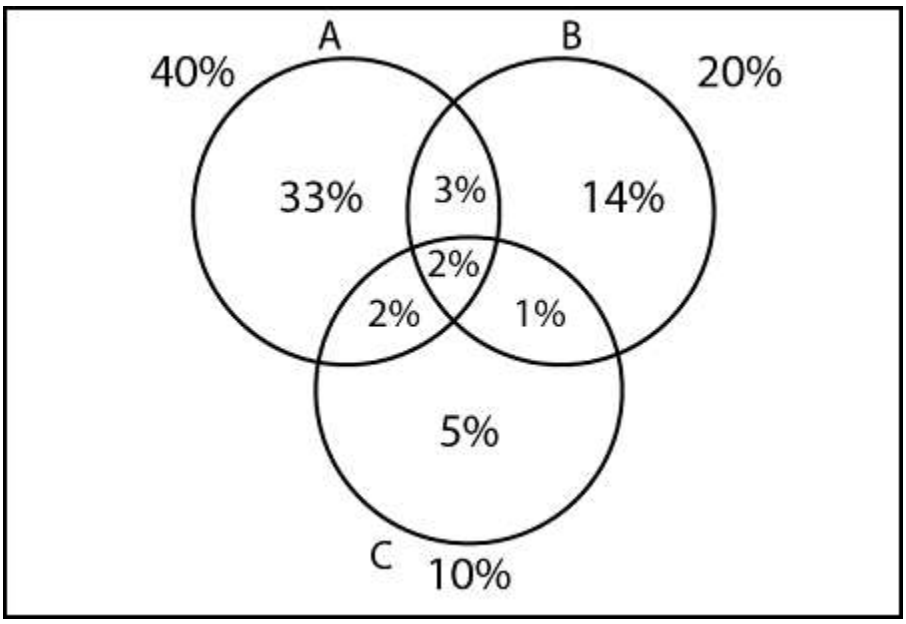
Example 2:

In a town of 10000 families it was found that 40% of families buy newspaper A, 20% family buy newspaper B, 10% family buy newspaper C, 5% family buy newspaper A and B, 3% family buy newspaper B and C and 4% family buy newspaper A and C. If 2% family buy all the newspaper. Find the number of families which buy

1. Number of families which buy all three newspapers.
2. Number of families which buy newspaper A only
3. Number of families which buy newspaper B only
4. Number of families which buy newspaper C only
5. Number of families which buy None of A, B, C
6. Number of families which buy exactly only one newspaper
7. Number of families which buy newspaper A and B only
8. Number of families which buy newspaper B and C only
9. Number of families which buy newspaper C and A only
10. Number of families which buy at least two newspapers

11. Number of families which buy at most two newspapers
12. Number of families which buy exactly two newspapers

Solution:



1. Number of families which buy all three newspapers:

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$
$n(A \cup B \cup C) = 40 + 20 + 10 - 5 - 3 - 4 + 2 = 60\%$

2. Number of families which buy newspaper A only

$= 40 - 7 = 33\%$

3. Number of families which buy newspaper B only

$= 20 - 6 = 14\%$

4. Number of families which buy newspaper C only

$= 10 - 5 = 5\%$

5. Number of families which buy None of A, B, and C

$n\ (A \cup B \cup C)^c = 100 - n\ (A \cup B \cup C)$

$n\ (A \cup B \cup C)^c = 100 - [40 + 20 + 10 - 5 - 3 - 4 + 2]$

$n\ (A \cup B \cup C)^c = 100 - 60 = 40\ \%$

6. Number of families which buy exactly only one newspaper

= 33 + 14 + 5 = 52%

7. Number of families which buy newspaper A and B only

= 3%

8. Number of families which buy newspaper B and C only

= 1%

9. Number of families which buy newspaper C and A only

= 2%

10. Number of families which buy at least two newspapers

= 8%

11. Number of families which buy at most two newspapers

= 98%

12. Number of families which buy exactly two newspapers

= 6%

Computer Representation of Sets

Computer Representation of Sets

Method for storing elements using an arbitrary ordering of the elements of the universal set.Specify an arbitrary ordering of the elements of U, for instance $a1, a2, \ldots, an$. Represent a subset A of U with the bit string of length n, where the ith bit in this string is 1 if ai belongs to A and is 0 if ai does not belong to A.

Example1: Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

i)What bit strings represent the subset of all odd integers in U?

The bit string that represents the set of odd integers in U, $\{1, 3, 5, 7, 9\}$, has a one bit in the first, third, fifth, seventh, and ninth positions. It is 10 1010 1010.

ii)What bit strings represent the subset of all even integers in U?

The bit string that represent the subset of even integers in U,$\{2, 4, 6, 8, 10\}$.

It is 01 0101 0101.

iii)What bit strings represent the subset of integers not exceeding 5 in U?

The set of all integers in U that do not exceed 5, $\{1, 2, 3, 4, 5\}$, is represented by the string 11 1110 0000.

To find the bit string for the **complement** of a set from the bit string for that set, change each 1 to 0 and each 0 to 1.

Example 2: The bit string for the set $\{1, 3, 5, 7, 9\}$ (with universal set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$) is 10 1010 1010.

What is the bit string for the **complement** of this set?

The bit string for the **complement** of this set is obtained by replacing 0s with 1s.

This yields the string 01 0101 0101,which corresponds to the set $\{2, 4, 6, 8, 10\}$.

To obtain the bit string for the union and intersection of two sets we perform bitwise Boolean operations on the bit strings representing the two sets.

Example 3: The bit strings for the sets $\{1, 2, 3, 4, 5\}$ and $\{1, 3, 5, 7, 9\}$ are 11 1110 0000 and 10 1010 1010.

The bit string for the **union** of these sets is 11 1110 0000 $\vee$ 10 1010 1010 = 11 1110 1010,

which corresponds to the set {1, 2, 3, 4, 5, 7, 9}.

If either of the bits in the ith position in the two strings is 1 (or both are 1),the bit in the ith position of the bit string of the **union** is 1. When both bits are 0, is 0. Hence, the bit string for union is the bitwise OR of the bit strings for the two sets.

The bit string for the **intersection** of these sets is

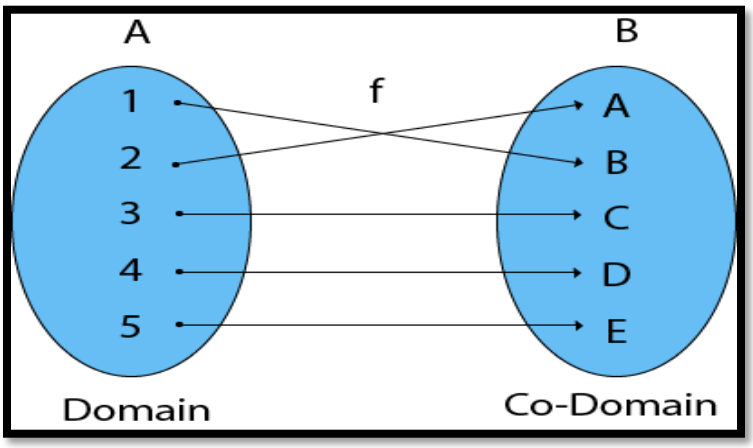
11 1110 0000 ∧ 10 1010 1010 = 10 1010 0000, which corresponds to the set {1, 3, 5}.

When the bits in the corresponding position in the two strings are both 1,the bit in the ith position of the bit string of the intersection is 1. When either of the two bits is 0 (or both are 0) ,is 0. Hence, the bit string for the intersection is the bitwise AND of the bit strings for the two sets.

Functions:

Functions

It is a mapping in which every element of set A is uniquely associated at the element with set B. The set of A is called Domain of a function and set of B is called Co domain.



Domain, Co-Domain, and Range of a Function:

Domain of a Function: Let f be a function from P to Q. The set P is called the domain of the function f.

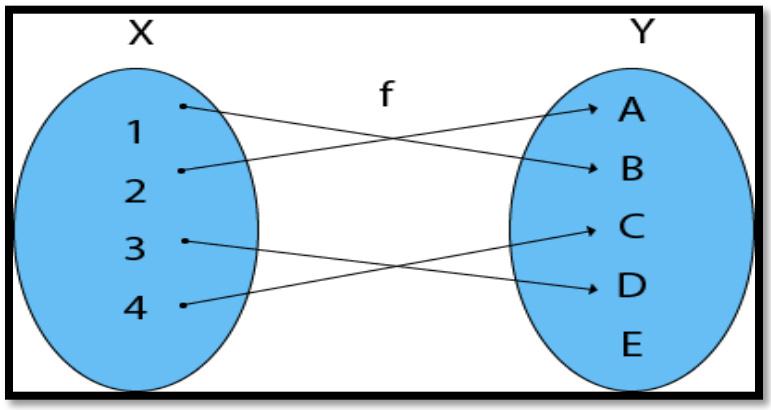
Co-Domain of a Function: Let f be a function from P to Q. The set Q is called Co-domain of the function f.

Range of a Function: The range of a function is the set of picture of its domain. In other words, we can say it is a subset of its co-domain. It is denoted as f (domain).

1. If f: P → Q, then f (P) = {f(x): x ∈ P} = {y: y ∈ Q | ∃ x ∈ P, such that f (x) = y}.

Example: Find the Domain, Co-Domain, and Range of function.

1. Let x = {1, 2, 3, 4}
2. y = {a, b, c, d, e}
3. f = {(1, b), (2, a), (3, d), (4, c)



Solution:

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Domain of function: {1, 2, 3, 4}
Range of function: {a, b, c, d}
Co-Domain of function: {a, b, c, d, e}
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Functions as a Set

If P and Q are two non-empty sets, then a function f from P to Q is a subset of P x Q, with two important restrictions

1. ∀ a ∈ P, (a, b) ∈ f for some b ∈ Q
2. If (a, b) ∈ f and (a, c) ∈ f then b = c.

Note1: There may be some elements of the Q which are not related to any element of set P.

2. Every element of P must be related with at least one element of Q.

Example1: If a set A has n elements, how many functions are there from A to A?

Solution: If a set A has n elements, then there are n^n functions from A to A.

Representation of a Function

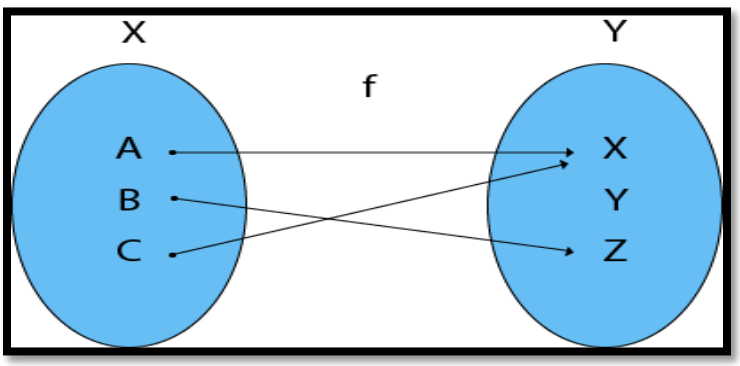
The two sets P and Q are represented by two circles. The function f: P → Q is represented by a collection of arrows joining the points which represent the elements of P and corresponds elements of Q

Example1:

Let X = {a, b, c} and Y = {x, y, z} and f: X → Y such that

f= {(a, x), (b, z), (c, x)}

Then f can be represented diagrammatically as follows



Example2: Let X = {x, y, z, k} and Y = {1, 2, 3, 4}. Determine which of the following functions. Give reasons if it is not. Find range if it is a function.

a. f = {(x, 1), (y, 2), (z, 3), (k, 4)
b. g = {(x, 1), (y, 1), (k, 4)
c. h = {(x, 1), (x, 2), (x, 3), (x, 4)
d. l = {(x, 1), (y, 1), (z, 1), (k, 1)}
e. d = {(x, 1), (y, 2), (y, 3), (z, 4), (z, 4)}.

Solution:

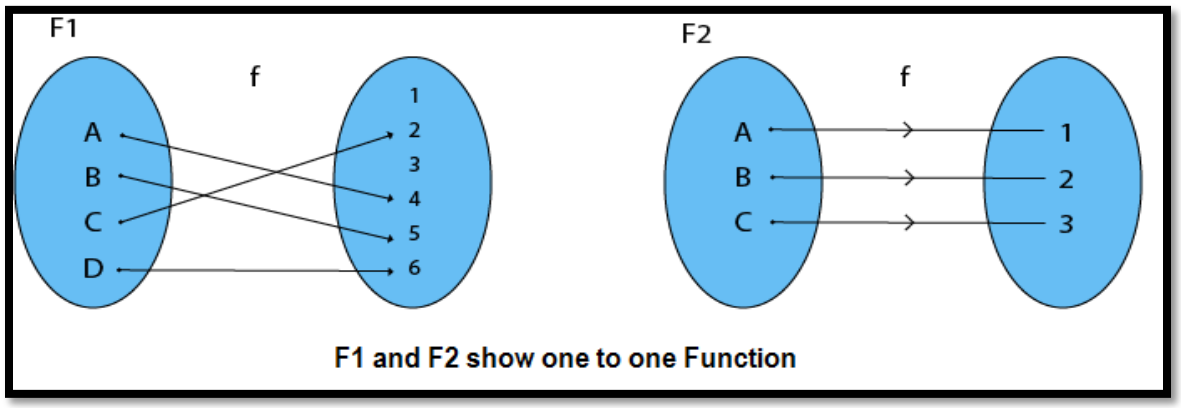
It is a function. Range (f) = {1, 2, 3, 4}

It is not a function because every element of X does not relate with some element of Y i.e., Z is not related with any element of Y.

h is not a function because h (x) = {1, 2, 3, 4} i.e., element x has more than one image in set Y.d is not a function because d (y) = {2, 3} i.e., element y has more than image in set Y.

Types of Functions

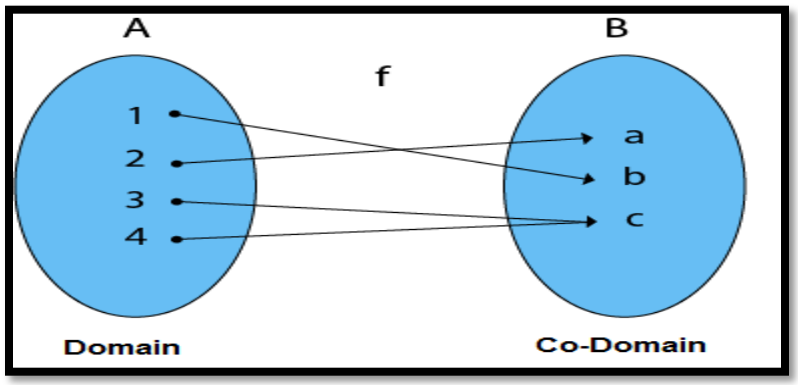
1. Injective (One-to-One) Functions: A function in which one element of Domain Set is connected to one element of Co-Domain Set.



2. Surjective (Onto) Functions: A function in which every element of Co-Domain Set has one pre-image.

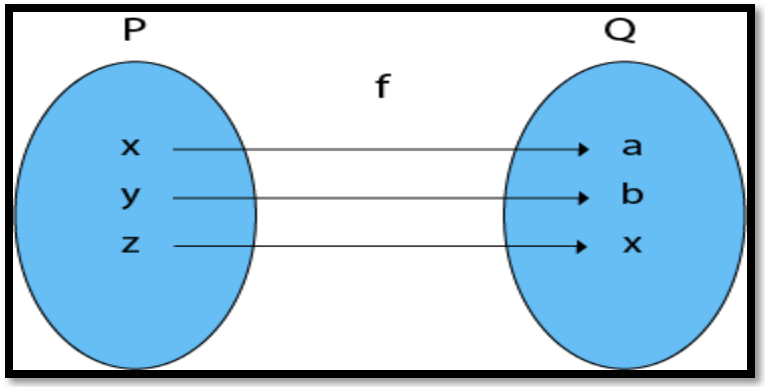
Example: Consider, A = {1, 2, 3, 4}, B = {a, b, c} and f = {(1, b), (2, a), (3, c), (4, c)}.

It is a Surjective Function, as every element of B is the image of some A



Note: In an Onto Function, Range is equal to Co-Domain.

3. Bijective (One-to-One Onto) Functions: A function which is both injective (one to - one) and surjective (onto) is called bijective (One-to-One Onto) Function.



Example:

Consider P = {x, y, z}

Q = {a, b, c}

and f: P → Q such that

f = {(x, a), (y, b), (z, c)}

The f is a one-to-one function and also it is onto. So it is a bijective function.

4. Into Functions: A function in which there must be an element of co-domain Y does not have a pre-image in domain X.

Example:

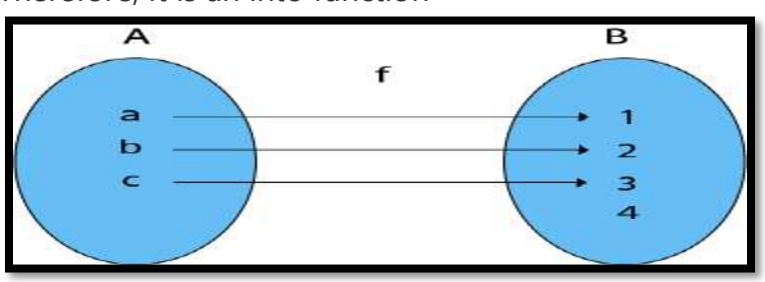
Consider, A = {a, b, c}

B = {1, 2, 3, 4} and f: A → B such that

f = {(a, 1), (b, 2), (c, 3)}

In the function f, the range i.e., {1, 2, 3} ≠ co-domain of Y i.e., {1, 2, 3, 4}

Therefore, it is an into function



Invertible (Inverse) Functions

A function f: X → Y is invertible if and only if it is a bijective function.

Consider the bijective (one to one onto) function f: X → Y. As f is a one to one, therefore, each element of X corresponds to a distinct element of Y. As f is onto, there is no element of Y which is not the image of any element of X, i.e., range = co-domain Y.

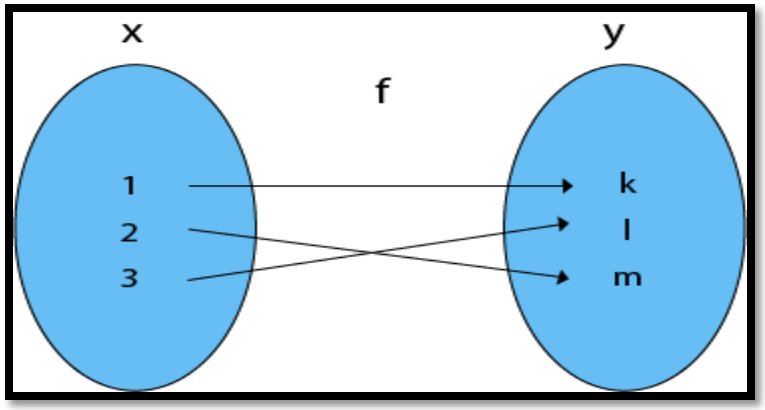
The inverse function for f exists if f^{-1} is a function from Y to X.

Example:

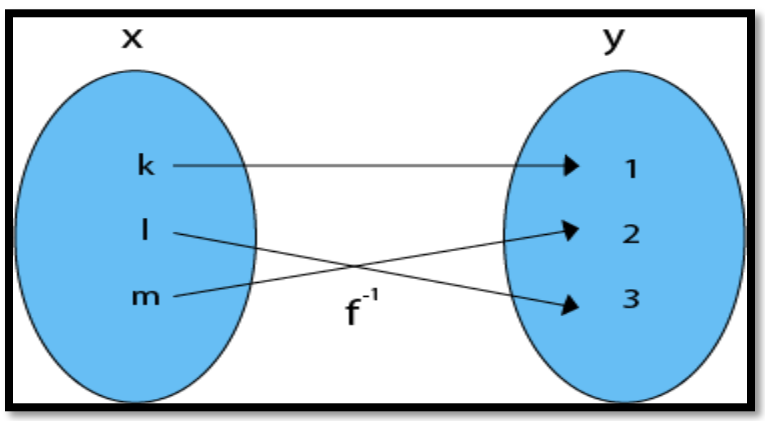
Consider, X = {1, 2, 3}

Y = {k, l, m} and f: X→Y such that

f = {(1, k), (2, m), (3, l)



The inverse function of f is shown in fig:



Compositions of Functions

Consider functions, f: A → B and g: B → C. The composition of f with g is a function from A into C defined by (gof) (x) = g [f(x)] and is defined by gof.

To find the composition of f and g, first find the image of x under f and then find the image of f (x) under g.

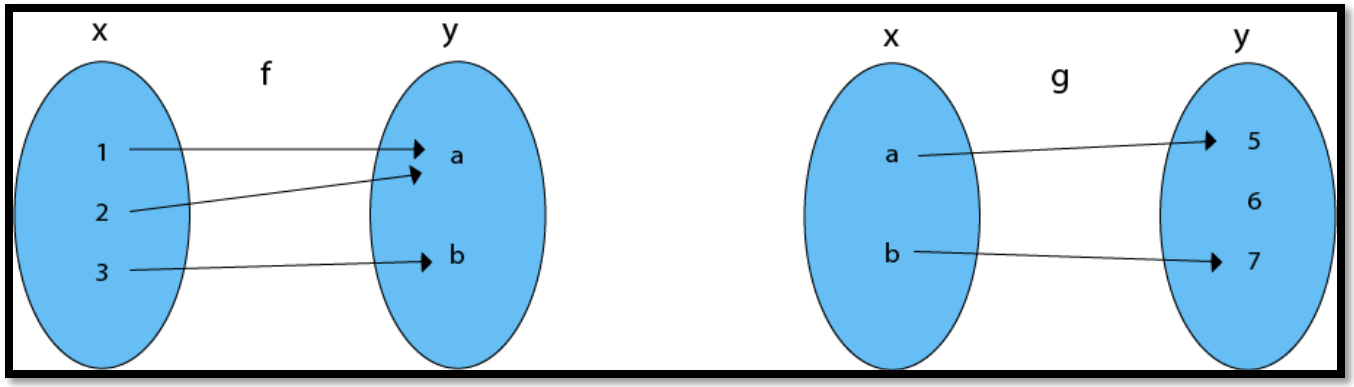
Example1:

Let X = {1, 2, 3}

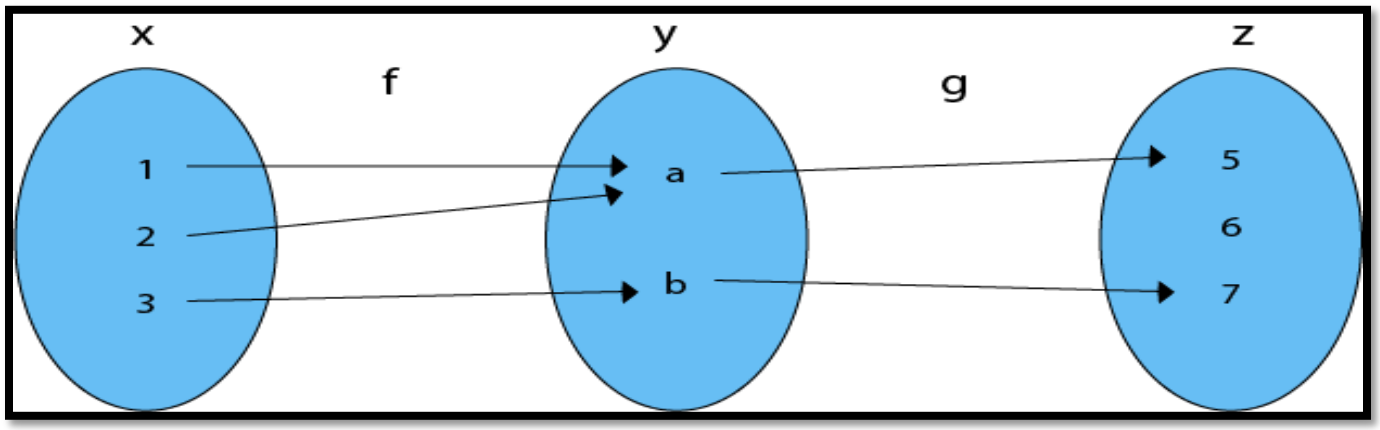
Y = {a, b}

Z = {5, 6, 7}.

Consider the function f = {(1, a), (2, a), (3, b)} and g = {(a, 5), (b, 7)} as in figure. Find the composition of gof.



Solution: The composition function gof is shown in fig:




```
(gof) (1) = g [f (1)] = g (a) = 5, (gof) (2) = g [f (2)] = g (a) = 5
(gof) (3) = g [f (3)] = g (b) = 7.
```

Example2: Consider f, g and h, all functions on the integers, by f (n) $=n^2$, g (n) = n + 1 and h (n) = n - 1.

Determine **(i)** hofog **(ii)** gofoh **(iii)** fogoh.

Solution:

```
(i) hofog (n) = n + 1,
    hofog (n + 1) = (n+1)²
h [(n+1)² ] = (n+1)² - 1 = n² + 1 + 2n - 1 = n² + 2n.

(ii) gofoh (n) = n - 1, gof (n - 1) = (n-1)²
     g [(n-1)² ] = (n-1)² + 1 = n² + 1 - 2n + 1 = n² - 2n + 2.

(iii) fogoh (n) = n - 1
      fog (n - 1) = (n - 1) + 1
      f (n) = n².
```

Note:

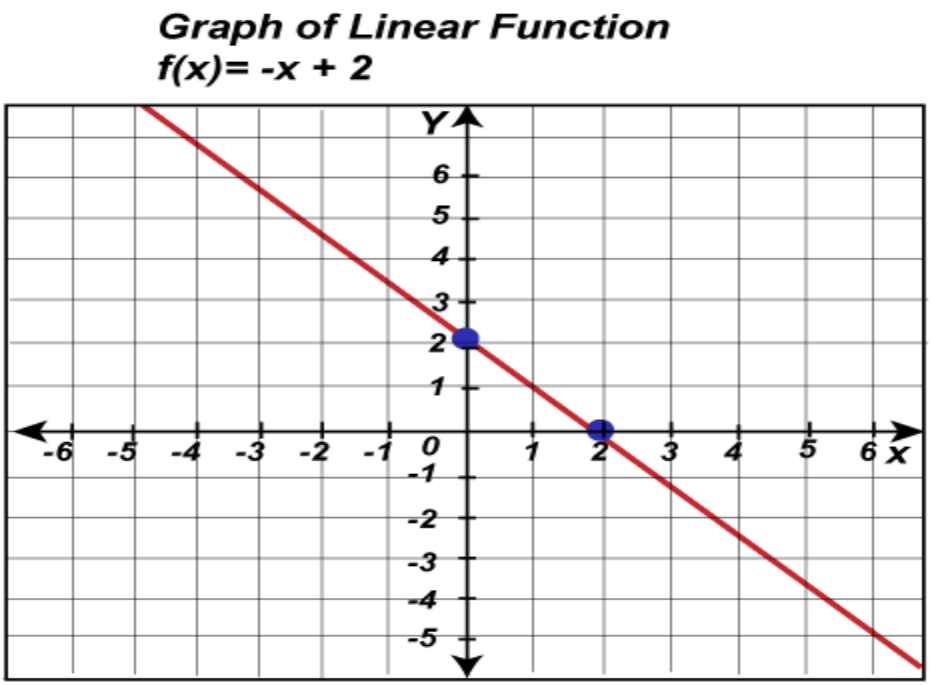
- If f and g are one-to-one, then the function (gof) (gof) is also one-to-one.
- If f and g are onto then the function (gof) (gof) is also onto.
- Composition consistently holds associative property but does not hold commutative property.

Graphing Functions

Graphing function can be described as a process of drawing the graph of a function. There are some simple functions and some complex functions in the basic graphing functions. The simple functions are cubic, linear, quadratic, and many more and the complex functions are logarithmic, rational, etc. In this section, we will understand the definition, basic functions, and examples of graphing functions.

Use of Graphing functions

With the help of a graphical function, we can draw a curve that is used to indicate the function on a coordinate plane. If this curve (graph) indicates a function, then each and every point on the curve will equally satisfy the function. For example: In this graph, we will show the linear function f(x) = -x+2.



Now we can take any point on the line from the above graph. Here we will take (-1, 3). Now we will substitute (-1, 3) = (x, y) in the function f(x) = -x+2. That means for this function, x = -1, and y = 3. We can also write the function f(x) = -x+2 in the form y = -x+2.

Now we will put the values of x and y in the function y = -x+2 and get the following:

3 = -(-1)+2

3 = 1+2
3 = 3
Hence, we can say that the point (-1, 3) satisfies the function.
Similarly, we can take different points from the above line and check whether those points satisfy the function. In this case, the function will be satisfied by every point on the line/curve. The process of drawing these types of curves, which represent the functions, is known as graphing functions.

1. Graphical Linear Functions
2. Graphical Quadratic functions
3. Graphical Complex Functions

Mathematical Functions(Functions for Computer Science)

The following are the functions which are widely used in computer science.

1. Floor Functions: The floor function for any real number x is defined as f (x) is the greatest integer 1 less than or equal to x. It is denoted by $\lfloor x \rfloor$.

Example: Determine the value of

(i)[3. 5] **(ii)**[-2.4] **(iii)**[3. 143].

Solution:

```
(i)[3 . 5]  = 3
(ii) [-2 .4] = -3
(iii) [3. 143] = 3
```

2. Ceiling Functions: The ceiling function for any real number x is defined as h (x) is the smallest integer greater than or equal to x. It is denoted by ⌈x⌉.

Example: Determine the value of

(i)[3. 5] **(ii)** [-2.4] **(iii)** [3. 143].

Solution:

```
(i)[3. 5] = 4
(ii) [-2 .4] = -2
(iii) [3. 143] = 4.
```

3. Remainder Functions: The integer remainder is obtained when some a is divided by m. It is denoted by a (MOD m). We can also define it as, a (MOD m) is the unique integer t such that a = Mq + t. Here q is quotient $0 \le r < M$.

Example: Determine the value of the following:

(i) 35 (MOD 7) **(ii)** 20 (MOD 3) **(iii)** 4 (MOD 9)

Solution:

```
(i) 35 (MOD 7) = 0
(ii) 20 (MOD 3) = 2
(iii) 4 (MOD 9) = 4
```

4. Exponential Functions: Consider two sets A and B. Let $A = B = I_+$ and also let $f: A \rightarrow B$ be defined by $f(n) = k^n$. Here n is a +ve integer. The function f is called the base k exponential function.

Note1: $k^t = k.\ k.\ k.....k$ (t times).

2: $k^0=1, k^{-M}=\frac{1}{k^M}$

3. For rational number, a/b, the exponential function is $k^{a/b} = \sqrt[b]{k^a} = (\sqrt[b]{k})^M$

Example: Determine the value of the following:

(i) 10^3 **(ii)** $5^{1/2}$ **(iii)** 3^{-5}

Solution:

$10^3 = 10.\ 10.\ 10 = 1000$

$5^{1/2} = 2.23607$

$3^{-5} = \frac{1}{3^5} = \frac{1}{3.3.3.3.3} = \frac{1}{243}$

5. Logarithmic Functions: Consider two sets A and B. Let A = B = R (the set of real numbers and also let f_n:A→B be defined for each positive integer n > 1 as $f_n(x) = \log_n(x)$ the base n of x.

Note1: $k = \log_n x$ and n^k are equivalent.
2. For any base n, $\log_n 1 = 0$ as $n^0 = 1$.
3. For any base n, $\log_n n = 1$ as $n^1 = n$.

Example: Determine the value of the following:

(i) $\log_2$?16 **(ii)** $\log_2$ 100 **(iii)** log_2 0.001.

Solution:

```
(i)log2 16 = 4 as 2^4=16.
(ii)log2 100 = 6 as 2^6= 64 but 2^7=128 which is greater
(iii)log2 0.001=-9 as 2^-9= 1/512 but 2^-10= 1/1024 which is greater.
```

Fuzzy:
Introduction:

- The word "fuzzy" means "vaguness (ambiguity)".
- Fuzziness occurs when the boundary of a piece of information is not clear-cut.
- Fuzzy sets - 1965 Lotfi Zadeh as an extension of classical notation set.
- Classical set theory allows the membership of the elements in the set in **binary terms**.
- Fuzzy set theory permits membership function valued in the interval [0,1].

S.No	Crisp Set	Fuzzy Set
1	Crisp set defines the value is either 0 or 1.	Fuzzy set defines the value between 0 and 1 including both 0 and 1.
2	It is also called a classical set.	It specifies the degree to which something is true.
3	It shows full membership	It shows partial membership.
4	Eg1. She is 18 years old. Eg2. Rahul is 1.6m tall	Eg1. She is about 18 years old. Eg2. Rahul is about 1.6m tall.
5	Crisp set application used for digital design.	Fuzzy set used in the fuzzy controller.
6	It is bi-valued function logic.	It is infinite valued function logic
7	Full membership means totally true/false, yes/no, 0/1.	Partial membership means true to false, yes to no, 0 to 1.

<u>Fuzzy Sets</u>

• Fuzzy Logic is derived from fuzzy set theory
• Many degree of membership (between 0 to 1) are allowed.
• Thus a membership function μA(x) is associated with a fuzzy sets Ã such that the function maps every element of
universe of discourse X to the interval [0,1].
• The mapping is written as: $\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) | x \in X\}$

• Fuzzy Logic is capable of handing inherently imprecise (vague or inexact or rough or inaccurate) concepts
• Fuzzy set is defined as follows:
• If X is an universe of discourse and x is a particular element of X, then a fuzzy set A defined on X and can be written as a collection of ordered pairs
A = {(x, μÃ (x)), x є X }

Example
• Let X = {g1, g2, g3, g4, g5} be the reference set of students.
• Let Ã be the fuzzy set of "smart" students, where "smart" is
fuzzy term.
Ã = {(g1,0.4)(g2,0.5)(g3,1)(g4,0.9)(g5,0.8)}
Here Ã indicates that the smartness of g1 is 0.4 and so on

Membership Function
• The membership function fully defines the fuzzy set
• A membership function provides a measure of the degree of similarity of an element to a fuzzy set
Membership functions can
– either be chosen by the user arbitrarily, based on the user's experience (MF chosen by two users could be different depending upon their experiences,
perspectives, etc.)
– Or be designed using machine learning methods (e.g., artificial neural networks, genetic algorithms, etc.)

There are different shapes of membership functions;
• Triangular,
• Trapezoidal,
• Gaussian, etc

Operations on Fuzzy Set

Given Ã and B are the two fuzzy sets, and X be the universe of discourse with the following respective member functions:

$\mu_{\tilde{A}}(x)$ and $\mu_{\tilde{B}}(x)$

The operations of Fuzzy set are as follows:

1. Union Operation: The union operation of a fuzzy set is defined by:

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

Example:

Let's suppose A is a set which contains following elements:

$A = \{(X_1, 0.6), (X_2, 0.2), (X_3, 1), (X_4, 0.4)\}$

And, B is a set which contains following elements:

$B = \{(X_1, 0.1), (X_2, 0.8), (X_3, 0), (X_4, 0.9)\}$

then,

$AUB = \{(X_1, 0.6), (X_2, 0.8), (X_3, 1), (X_4, 0.9)\}$

Because, according to this operation

For X_1

$\mu_{A\cup B}(X_1) = \max (\mu_A(X_1), \mu_B(X_1))$

$\mu_{A\cup B}(X_1) = \max (0.6, 0.1)$

$\mu_{A\cup B}(X_1) = 0.6$

For X_2

$\mu_{A\cup B}(X_2) = \max (\mu_A(X_2), \mu_B(X_2))$

$\mu_{A\cup B}(X_2) = \max (0.2, 0.8)$

$\mu_{A\cup B}(X_2) = 0.8$

For X_3

$\mu_{A\cup B}(X_3) = \max (\mu_A(X_3), \mu_B(X_3))$

$\mu_{A\cup B}(X_3) = \max (1, 0)$

$\mu_{A\cup B}(X_3) = 1$

For X_4

$\mu_{A\cup B}(X_4) = \max (\mu_A(X_4), \mu_B(X_4))$

$\mu_{A\cup B}(X_4) = \max (0.4, 0.9)$

$\mu_{A\cup B}(X_4) = 0.9$

2. Intersection Operation: The intersection operation of fuzzy set is defined by:

$\mu_{A\cap B}(x) = \min (\mu_A(x), \mu_B(x))$

Example:

Let's suppose A is a set which contains following elements:

$A = \{(X_1, 0.3), (X_2, 0.7), (X_3, 0.5), (X_4, 0.1)\}$

And, B is a set which contains following elements:

$B = \{(X_1, 0.8), (X_2, 0.2), (X_3, 0.4), (X_4, 0.9)\}$

then,

$A\cap B = \{(X_1, 0.3), (X_2, 0.2), (X_3, 0.4), (X_4, 0.1)\}$

Because, according to this operation

For X_1

$\mu_{A\cap B}(X_1) = \min (\mu_A(X_1), \mu_B(X_1))$

$\mu_{A\cap B}(X_1) = \min (0.3, 0.8)$

$\mu_{A\cap B}(X_1) = 0.3$

For X_2

$\mu_{A\cap B}(X_2) = \min(\mu_A(X_2), \mu_B(X_2))$

$\mu_{A\cap B}(X_2) = \min(0.7, 0.2)$

$\mu_{A\cap B}(X_2) = 0.2$

For X_3

$\mu_{A\cap B}(X_3) = \min(\mu_A(X_3), \mu_B(X_3))$

$\mu_{A\cap B}(X_3) = \min(0.5, 0.4)$

$\mu_{A\cap B}(X_3) = 0.4$

For X_4

$\mu_{A\cap B}(X_4) = \min(\mu_A(X_4), \mu_B(X_4))$

$\mu_{A\cap B}(X_4) = \min(0.1, 0.9)$

$\mu_{A\cap B}(X_4) = 0.1$

3. Complement Operation: The complement operation of fuzzy set is defined by:

$\mu_{\bar{A}}(x) = 1-\mu_A(x)$,

Example:

Let's suppose A is a set which contains following elements:

$A = \{(X_1, 0.3), (X_2, 0.8), (X_3, 0.5), (X_4, 0.1)\}$

then,

$\bar{A} = \{(X_1, 0.7), (X_2, 0.2), (X_3, 0.5), (X_4, 0.9)\}$

Because, according to this operation

For X_1

$\mu_{\bar{A}}(X_1) = 1-\mu_A(X_1)$

$\mu_{\bar{A}}(X_1) = 1 - 0.3$

$\mu_{\bar{A}}(X_1) = 0.7$

For X_2

$\mu_{\bar{A}}(X_2) = 1-\mu_A(X_2)$

$\mu_{\bar{A}}(X_2) = 1 - 0.8$

$\mu_{\bar{A}}(X_2) = 0.2$

For X_3

$\mu_{\bar{A}}(X_3) = 1-\mu_A(X_3)$

$\mu_{\bar{A}}(X_3) = 1 - 0.5$

$\mu_{\bar{A}}(X_3) = 0.5$

For X_4

$\mu_{\bar{A}}(X_4) = 1-\mu_A(X_4)$

$\mu_{\bar{A}}(X_4) = 1 - 0.1$

$\mu_{\bar{A}}(X_4) = 0.9$

Sequence and Series Definition

A sequence is an arrangement of any objects or a set of numbers in a particular order followed by some rule. If a_1, a_2, a_3, a_4,…… etc. denote the terms of a sequence, then 1,2,3,4,…..denotes the position of the term.

A sequence can be defined based on the number of terms i.e. either finite sequence or infinite sequence.

If a_1, a_2, a_3, a_4, ……. is a sequence, then the corresponding series is given by

$S_N = a_1+a_2+a_3 + .. + a_N$

Note: The series is finite or infinite depending if the sequence is finite or infinite.

Types of Sequence and Series

Some of the most common examples of sequences are:

- Arithmetic Sequences
- Geometric Sequences
- Harmonic Sequences
- Fibonacci Numbers

Arithmetic Sequences

A sequence in which every term is created by adding or subtracting a definite number to the preceding number is an arithmetic sequence.

Geometric Sequences

A sequence in which every term is obtained by multiplying or dividing a definite number with the preceding number is known as a geometric sequence.

Harmonic Sequences

A series of numbers is said to be in harmonic sequence if the reciprocals of all the elements of the sequence form an arithmetic sequence.

Fibonacci Numbers

Fibonacci numbers form an interesting sequence of numbers in which each element is obtained by adding two preceding elements and the sequence starts with 0 and 1. Sequence is defined as, $F_0 = 0$ and $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$

Sequence and Series Formulas

List of some basic formula of arithmetic progression and geometric progression are

	Arithmetic Progression	**Geometric Progression**
Sequence	a, a+d, a+2d,……,a+(n-1)d,….	a, ar, ar^2,….,$ar^{(n-1)}$,…
Common Difference or Ratio	Successive term – Preceding term Common difference = d = $a_2 - a_1$	Successive term/Preceding term Common ratio = r = $ar^{(n-1)}/ar^{(n-2)}$
General Term (nth Term)	$a_n = a + (n-1)d$	$a_n = ar^{(n-1)}$
nth term from the last term	$a_n = l - (n-1)d$	$a_n = l/r^{(n-1)}$
Sum of first n terms	$s_n = n/2(2a + (n-1)d)$	$s_n = a(1 - r^n)/(1 - r)$ if $\|r\| < 1$ $s_n = a(r^n - 1)/(r - 1)$ if $\|r\| > 1$

*Here, a = first term, d = common difference, r = common ratio, n = position of term, l = last term

Difference Between Sequences and Series

Let us find out how a sequence can be differentiated with series.

Sequences	**Series**
Set of elements that follow a pattern	Sum of elements of the sequence
Order of elements is important	Order of elements is not so important
Finite sequence: 1,2,3,4,5	Finite series: 1+2+3+4+5
Infinite sequence: 1,2,3,4,……	Infinite Series: 1+2+3+4+……

Sequence and Series Examples

Question 1: **If 4,7,10,13,16,19,22……is a sequence, Find:**

1. **Common difference**
2. **nth term**
3. **21st term**

Solution: Given sequence is, 4,7,10,13,16,19,22……

a) The common difference = **7 – 4 = 3**

b) The nth term of the arithmetic sequence is denoted by the term T_n and is given by $T_n = a + (n-1)d$, where "a" is the first term and d is the common difference.
$T_n = 4 + (n – 1)3 = 4 + 3n – 3 = 3n + 1$
c) 21st term as: $\mathbf{T_{21} = 4 + (21-1)3 = 4+60 = 64}$.

Question 2: Consider the sequence 1, 4, 16, 64, 256, 1024….. Find the common ratio and 9th term.

Solution: The common ratio (r) = 4/1 = 4

The preceding term is multiplied by 4 to obtain the next term.

The nth term of the geometric sequence is denoted by the term T_n and is given by $T_n = ar^{(n-1)}$ where a is the first term and r is the common ratio.

Here a = 1, r = 4 and n = 9

So, 9th term is can be calculated as $T_9 = 1* (4)^{(9-1)}= 4^8 = 65536$.

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<u>Recurrence Relations</u>

Definition : A recurrence relation for the sequence {an} is an equation that expresses an in terms of one or more of the previous terms of the sequence, namely, a0, a1, … , an−1, for all integers n with n ≥ n0, where n0 is a nonnegative integer.
A sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation. (A recurrence relation is said to recursively define a sequence
EXAMPLE 1: Let {an} be a sequence that satisfies the recurrence relation an = an−1 + 3 for n = 1, 2, 3, … ,and suppose that a0 = 2. What are a1, a2, and a3?
Solution: We see from the recurrence relation that a1 = a0 + 3 = 2 + 3 = 5. It then follows that a2 = 5 + 3 = 8 and a3 = 8 + 3 = 11. ◂
EXAMPLE 2 Let {an} be a sequence that satisfies the recurrence relation an = an−1 − an−2 for n = 2, 3, 4, … ,and suppose that a0 = 3 and a1 = 5. What are a2 and a3?
Solution: We see from the recurrence relation that a2 = a1 − a0 = 5 − 3 = 2 and a3 = a2 − a1 =2 − 5 = −3. We can find a4, a5, and each successive term in a similar way.
Definition: The Fibonacci sequence, f0, f1, f2, … , is defined by the initial conditions f0 = 0, f1 = 1, andthe recurrence relation fn = fn−1 + fn−2 for n = 2, 3, 4, … .
EXAMPLE1: Find the Fibonacci numbers f2, f3, f4, f5, and f6.
Solution: The recurrence relation for the Fibonacci sequence tells us that we find successive

terms by adding the previous two terms. Because the initial conditions tell us that f0 = 0 and
f1 = 1, using the recurrence relation in the definition we find that
f2 = f1 + f0 = 1 + 0 = 1,
f3 = f2 + f1 = 1 + 1 = 2,
f4 = f3 + f2 = 2 + 1 = 3,
f5 = f4 + f3 = 3 + 2 = 5,
f6 = f5 + f4 = 5 + 3 = 8. ◂

Summations:

Are simply the sums of the terms in a sequence.
Some people call them "series" instead of summations (I've never figured out why).I.e., a summation is, by definition, the sum of some or all terms in a sequence.
So, given the sequence

$$a_1, a_2, \ldots, a_n, \ldots,$$

the corresponding summation is

$$a_1 + a_2 + \ldots + a_n + \ldots$$

You'd think that this would have to be a very large number (after all, you keep adding more and more numbers),and it often *is* quite large, but sometimes such a summation "converges" to a finite number and turns out to be a convenient way to compute the number that is the sum.

For instance, consider this sequence:

$$1, -1/3, 1/5, -1/7, 1/9, \ldots$$

where:

if Odd(n), then $a_n = +1/(2n-1)$
else $a_n = -1/(2n-1)$

Check it out:

Odd(1), so $a_1 = 1/(2*1 - 1) = 1/(2-1) = 1/1 = 1$

Even(2), so $a_2 = -1/(2*2 - 1) = -1/(4-1) = -1/3$

etc.

Now consider its summation:

$$1 + -1/3 + 1/5 + -1/7 + 1/9 + \ldots$$

$$= 1 - 1/3 + 1/5 - 1/7 + 1/9 - \ldots$$

As you might guess, this converges (i.e., it does not get infinitely large).
After all, each time we're adding on a *smaller* fraction, and some of them are negative, so we're actually subtracting them.
All of that will surely make each successive "partial" sum only a very little bit bigger than the previous some, and in some cases, it'll be smaller.

Let's compute the partial sums:

The sum of the first term is just 1.
The sum of the first 2 terms is 1 – 1/3 = 2/3 = 0.6666666666666666666…
The sum of the first 3 terms is 1 – 1/3 + 1/5 = 0.66… + 0.2 = 0.8666…
The sum of the first 4 terms is 1 – 1/3 + 1/5 – 1/7 = 0.866… – 0.1428571… = 0.723809523…
The sum of the first 5 terms is 0.723809523… + 1/9 = 0.723809523… + 0.11111… = 0.834920634…etc.

It turns out (and I *won't* ask you to prove this!) that the sum of the infinite sequence is (get ready… drum roll, please):

$\pi/4$ (!)

(Consider what happens when you multiply each term in the summation by 4:

$$4 - 4/3 + 4/5 - 4/7 + 4/9 - \ldots$$

The partial sum of these first 5 terms is 3.33968253…; it's getting close to π.)

So, you could write a computer program to compute π to any degree of accuracy(i.e., to any position in the decimal expansion of π)by writing a for-loop (or a "count-loop") that adds up these numbers.

It doesn't always happen that a series like this converges. Consider:

$$1 + 1/2 + 1/3 + 1/4 + 1/5 + \ldots$$

This diverges!
(I.e., it grows infinitely large, despite the fact that each number you add on is smaller than the previous one! It's just that it's not small enough to make the summation converge.)

The final point about summations that I want to make is the notation for it.
The symbol for a summation is the capital Greek letter sigma,
which kind of looks like a backwards "3" with angles instead of curves:

$$\Sigma$$

Just as with the big union and big intersection symbols, we can use this as a shorthand;
so, instead of writing:

$$a_1 + a_2 + a_3 + a_4 + \ldots$$

we'll write:

$$\Sigma\, a_n$$

But we have to say where n begins and where it "ends".
Typically, for an infinite summation, it won't end,
so we use the infinity sign, which looks like the numeral "8" lying on its side:

$$\infty$$

Below the Σ, we write something like "n=1" or "n=0" to indicate where we start adding,depending on whether the range of the sequence is $\mathbf{Z}^+$ or $\mathbf{N}$,
and above the Σ we write either "n=k", if we want to stop at the kth term, or "∞" if we want the infinite sum.
So the summation above could be written something like:

$$\sum_{n=1}^{\infty} a_n$$

or, if we write it like this:

$$\Sigma n \in \mathbf{Z}^+ (a_n)$$

it kind of looks like a quantifier expression with a bound variable.

It is essentially what programming languages call a "for-loop" or a "count-loop":
For $n = 1$ to infinity, or for $n = 1$ to k, add up the numbers or the output of the function (i.e., the numbers in the sequence).

What is the value of $\sum_{j=1}^{5} j^2$?

Solution: We have

$$\begin{aligned}\sum_{j=1}^{5} j^2 &= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 \\ &= 1 + 4 + 9 + 16 + 25 \\ &= 55.\end{aligned}$$

Double summations arise in many contexts (as in the analysis of nested loops in computer programs). An example of a double summation is

$$\sum_{i=1}^{4} \sum_{j=1}^{3} ij.$$

To evaluate the double sum, first expand the inner summation and then continue by computing the outer summation:

$$\begin{aligned}\sum_{i=1}^{4} \sum_{j=1}^{3} ij &= \sum_{i=1}^{4} (i + 2i + 3i) \\ &= \sum_{i=1}^{4} 6i \\ &= 6 + 12 + 18 + 24 = 60.\end{aligned}$$

◀

We can also use summation notation to add all values of a function, or terms of an indexed set, where the index of summation runs over all values in a set. That is, we write

$$\sum_{s \in S} f(s)$$

to represent the sum of the values $f(s)$, for all members s of S.

What is the value of $\sum_{s \in \{0,2,4\}} s$?

Solution: Because $\sum_{s \in \{0,2,4\}} s$ represents the sum of the values of s for all the members of the set $\{0, 2, 4\}$, it follows that

$$\sum_{s \in \{0,2,4\}} s = 0 + 2 + 4 = 6.$$

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TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x\rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, \lvert x\rvert < 1$	$\frac{1}{(1-x)^2}$

(*Requires calculus*) Let x be a real number with $|x| < 1$. Find $\sum_{n=0}^{\infty} x^n$.

Solution: By Theorem 1 with $a = 1$ and $r = x$ we see that $\sum_{n=0}^{k} x^n = \frac{x^{k+1} - 1}{x - 1}$. Because $|x| < 1$, x^{k+1} approaches 0 as k approaches infinity. It follows that

$$\sum_{n=0}^{\infty} x^n = \lim_{k \to \infty} \frac{x^{k+1} - 1}{x - 1} = \frac{0 - 1}{x - 1} = \frac{1}{1 - x}.$$

◀

Lab Work

Lab 1

Write a C program to find union of two sets.

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
```

```
int A[] = {1,2,3,4,5,6,7,8};
int B[] = {2,4,6,8,10,12};
int C[14];
int i,j,k=0,t;
for(i=0;i<8;i++)
{
   C[k++]=A[i];
}
for(i=0;i<6;i++)
{
   t=0;
   for(j=0;j<8;j++)
   {
      if(B[i]==A[j])
      {
         t=1;
         break;
      }
   }
   if(t==0)
   {
      C[k++]=B[i];
   }
}
printf("Set A = {");
for(i=0;i<8;i++)
{
  printf("%d",A[i]);
  if(i==7)
  continue;
  printf(",");
}
printf("}\n");
printf("Set B = {");
for(i=0;i<5;i++)
{
  printf("%d",B[i]);
  if(i==4)
  continue;
  printf(",");
}
printf("}\n");
printf("The Union set C = {");
for(i=0;i<k;i++)
```

```
  {
    printf("%d",C[i]);
    if(i==k-1)
    continue;
    printf(",");
  }
  printf("}\n");
   getch();
   return 0;
}
```

Output:

S et A = {1,2,3,4,5,6,7,8}

Set B = {2,4,6,8,10}

The Union set C = {1,2,3,4,5,6,7,8,10,12}

Write a C program to find intersection of sets

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
  int A[] = {1,2,3,4,5,6,7,8};
  int B[] = {2,4,6,8,10};
  int C[5];
  int i,j,k=0;
  for(i=0;i<8;i++)
  {
     for(j=0;j<5;j++)
     {
       if(A[i]==B[j])
       {
           C[k++] = A[i];
       }
     }
  }
  printf("Set A = {");
  for(i=0;i<8;i++)
  {
    printf("%d",A[i]);
    if(i==7)
    continue;
    printf(",");
  }
  printf("}\n");
```

```
  printf("Set B = {");
  for(i=0;i<5;i++)
  {
    printf("%d",B[i]);
    if(i==4)
    continue;
    printf(",");
  }
  printf("}\n");
  printf("The intersection set C = {");
  for(i=0;i<k;i++)
  {
    printf("%d",C[i]);
    if(i==k-1)
    continue;
    printf(",");
  }
  printf("}\n");
   getch();
   return 0;
}
```

Output

Set A = {1,2,3,4,5,6,7,8}
Set B = {2,4,6,8,10}
The intersection set C = {2,4,6,8}

Write a C program to find difference of two sets

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
  int A[] = {1,2,3,4,5,6,7,8};
  int B[] = {2,4,6,8,10};
  int C[5];
  int i,j,k=0,t;
  for(i=0;i<8;i++)
  {
    t=0;
      for(j=0;j<5;j++)
     {
       if(A[i]==B[j])
       {
          t=1;
```

```
        }
      }
      if(t==0)
      C[k++]=A[i];
  }
  printf("Set A = {");
  for(i=0;i<8;i++)
  {
    printf("%d",A[i]);
    if(i==7)
    continue;
    printf(",");
  }
  printf("}\n");
  printf("Set B = {");
  for(i=0;i<5;i++)
  {
    printf("%d",B[i]);
    if(i==4)
    continue;
    printf(",");
  }
  printf("}\n");
  printf("The set C= A-B = {");
  for(i=0;i<k;i++)
  {
    printf("%d",C[i]);
    if(i==k-1)
    continue;
    printf(",");
  }
  printf("}\n");
   getch();
   return 0;
}
```

Output:

Set A = {1,2,3,4,5,6,7,8}

Set B = {2,4,6,8,10}

The set C= A-B = {1,3,5,7}

Write a C program to find Cartesian product of two sets.

Ans:

```
#include<stdio.h>
#include<conio.h>
```

```
#include<math.h>
int main()
{
  int A[] = {1,2,3,4,5,6,7,8};
  int B[] = {2,4,6,8,10};
  int C[5];
  int i,j,k=0,t;
  printf("Set A = {");
  for(i=0;i<8;i++)
  {
    printf("%d",A[i]);
    if(i==7)
    continue;
    printf(",");
  }
  printf("}\n");
  printf("Set B = {");
  for(i=0;i<5;i++)
  {
    printf("%d",B[i]);
    if(i==4)
    continue;
    printf(",");
  }
  printf("}\n");
  printf("The cartesian product of A and B = {");
  for(i=0;i<8;i++)
  {
     for(j=0;j<5;j++)
    {
      printf("(%d,%d),",A[i],B[j]);
    }

  }
  printf("}\n");
   getch();
   return 0;
}
```

Output:

Set A = {1,2,3,4,5,6,7,8}
Set B = {2,4,6,8,10}
The Cartesian product of A and B = {(1,2),(1,4),(1,6),(1,8),(1,10),(2,2),(2,4),(2,6),(2,8),(2,10),(3,2),(3,4),(3,6),(3,8),(3,10),(4,2),(4,4),(4,6),(4,8),(4,10),(5,2),(5,4),(5,6),(5,8),(5,10),(6,2),(6,4),(6,6),(6,8),(6,10),(7,2),(7,4),(7,6),

(7,8),(7,10),(8,2),(8,4),(8,6),(8,8),(8,10) }

Write a C program to find ceiling and floor value of a variable or expression

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
   float x = 2.456;
   float c = ceil(x);
   float f = floor(x);
   printf("The floor value of %f is = %f\n",x,f);
   printf("The ceiling value of %f is = %f\n",x,c);
   getch();
   return 0;
}
```

output:

The floor value of 2.456000 is = 2.000000

The ceiling value of 2.456000 is = 3.000000

Write a C program to find to implement fuzzy set operations

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
float min(float a,float b);
float max(float a,float b);
int main()
{
    float fa,fb,fi,fu,fac;
    float x = 1.0;
   printf("Enter membership function of first set:\n");
   scanf("%f",&fa);
   printf("Enter membership function of second set:\n");
   scanf("%f",&fb);
   fi = min(fa,fb);
   fu = max(fa,fb);
   fac = x-fa;
   printf("The membership function of intersection = %0.1f\n",fi);
   printf("The membership function of union = %0.1f\n",fu);
   printf("The membership function of complement of first set = %0.1f\n",fac);
   getch();
   return 0;
}
```

```
float min(float a, float b)
{
  if(a<b)
  return a;
  else
  return b;
}
 float max(float a, float b)
{
  if(a>b)
  return a;
  else
  return b;
}
```

Output:

```
Enter membership function of first set:
0.2
Enter membership function of second set:
0.4
The membership function of intersection = 0.2
The membership function of union = 0.4
The membership function of complement of first set = 0.8
```

Program to find union and intersection of two sets using bit string representation

```
#include<stdio.h>
#include<conio.h>
#define N 10
int main()
{
   int i,k=0,j,t;
   int U[N]= {1,2,3,4,5,6,7,8,9,10};
   int A[N] = {1,3,4,5,7};
   int B[N] = {1,2,4,6,8,10};
   int SA[N];
   int SB[N];
   int US[N];
   int CS[N];
   int DS[N];
   int C[N];
   int D[N];
   for(i=0;i<N;i++)
   {
```

```
  if(U[i]!=-1)
  US[i]=1;
}
printf("The Bit String representation of Universal set\n");
for(i=0;i<N;i++)
printf("%d ",US[i]);
printf("\n");
//Finding bit string for set A
for(i=0;i<N;i++)
{
   t=0;
   for(j=0;j<N;j++)
   {
       if(A[j]==U[i])
       {
         t=1;
         break;
       }
   }
   if(t==1)
   SA[i]=1;
   else
   SA[i]=0;
}
printf("The Bit String representation of set A\n");
for(i=0;i<N;i++)
printf("%d ",SA[i]);
printf("\n");

for(i=0;i<N;i++)
{
   t=0;
   for(j=0;j<N;j++)
   {
       if(B[j]==U[i])
       {
         t=1;
         break;
       }
   }
   if(t==1)
```

```
   SB[i]=1;
   else
   SB[i]=0;
}
//Finding bit string for set B
printf("The Bit String representation of set B\n");
for(i=0;i<N;i++)
printf("%d ",SB[i]);
printf("\n");
//computing union
for(i=0;i<10;i++)
{
     if(SA[i]==1||SB[i]==1)
     CS[i]=1;
     else
     CS[i]=0;
}
for(i=0;i<10;i++)
{
  if(CS[i]==1)
  C[k++]=U[i];
}
//computing intersection
for(i=0;i<10;i++)
{
     if(SA[i]==1&&SB[i]==1)
     DS[i]=1;
     else
     DS[i]=0;
}
printf("The union set is\n");
for(i=0;i<k;i++)
printf("%d ",C[i]);
k=0;
for(i=0;i<10;i++)
{
  if(DS[i]==1)
  D[k++]=U[i];
}
printf("\nThe intersection set is\n");
for(i=0;i<k;i++)
```

```
    printf("%d ",D[i]);
    getch();
    return 0;
}
```

Output:

The Bit String representation of Universal set

1 1 1 1 1 1 1 1 1 1

The Bit String representation of set A

1 0 1 1 1 0 1 0 0 0

The Bit String representation of set B

1 1 0 1 0 1 0 1 0 1

The union set is

1 2 3 4 5 6 7 8 10

The intersection set is

1 4

Assignment 1

1. State and prove De Morgan's law in set theory
2. Show that if A and B are set then prove that A-B = A∩B'
3. Draw Venn Diagrams for the following set: (A∩B)U(A∩C)
4. How sets are represented by using bit strings? Why it is preferred over unordered representation of sets.
5. Let f(x) = ax+b and g(x) = cx+d where a,b,c and d are constants. Determine for which constants a,b,c and d it is true that f o g = g o f
6. Show that a function f(x) = ax+b from R to R is invertible where a and b are constants with a≠0 and find the inverse of f.
7. How can you relate domain and co-domain of functions with functions in programming language? Discuss composite and inverse of function with suitable function with examples
8. State inclusion-exclusion principle in set theory. A total of 1232 students have taken a course in Spanish, 879 have taken a course in French and 114 have taken course in Russian. Further, 103 have taken courses in both Spanish and French. 23 have taken courses in both Spanish and Russian and 14 have taken courses in both French and

Russian. If 2092 students have taken at least one of Spanish, French and Russian, how many students have taken a course in all three languages?

9. In a town of 10000 families it was found that 40% of families buy newspaper A, 20% family buy newspaper B, 10% family buy newspaper C, 5% family buy newspaper A and B, 3% family buy newspaper B and C and 4% family buy newspaper A and C. If 2% family buy all the newspaper. Find the number of families which buy

a. Number of families which buy all three newspapers.
b. Number of families which buy newspaper A only
c. Number of families which buy newspaper B only
d. Number of families which buy newspaper C only
e. Number of families which buy None of A, B, C
f. Number of families which buy exactly only one newspaper
g. Number of families which buy newspaper A and B only
h. Number of families which buy newspaper B and C only
i. Number of families which buy newspaper C and A only

10. Find a at least three different sequences beginning with the terms 3,5,7 whose terms are generated by a simple formula or rule.